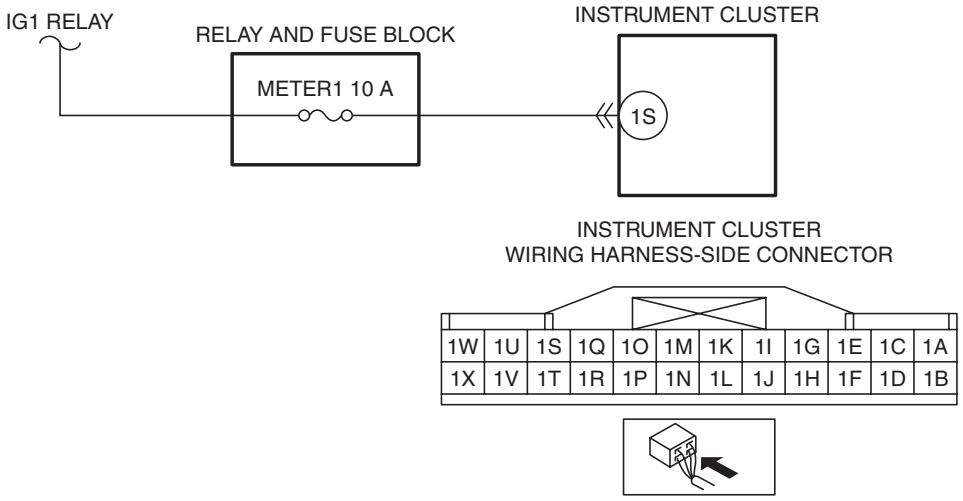
DTC U3003:16 [INSTRUMENT CLUSTER]

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Without i-stop System And Without i-ELOOP

Description	Low power supply voltage input to instrument cluster
Detection condition	• Instrument cluster power supply circuit voltage of **less than 10 V** is detected.
Fail-safe	Not applicable
Possible cause	• DTCs are stored in the PCM. • Battery malfunction • Generator malfunction • Instrument cluster connector or terminal malfunction • Instrument cluster power supply circuit malfunction — Short to ground in wiring harness between METER1 10 A fuse and instrument cluster terminal 1S — METER1 10 A fuse malfunction — Open circuit in wiring harness between IG1 relay and instrument cluster terminal 1S • Instrument cluster malfunction



Diagnostic Procedure

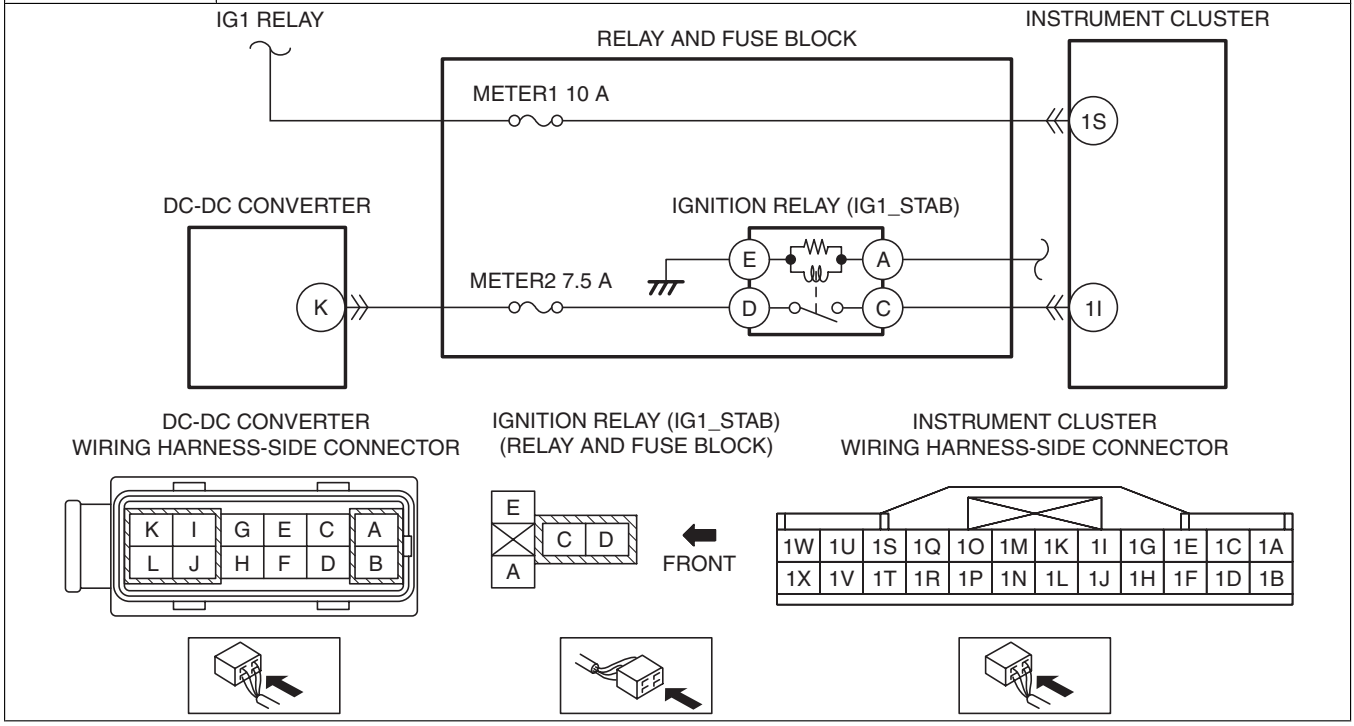
Step	Inspection		Action
1	**VERIFY PCM DTCs** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
2	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is the battery normal?	Yes	Go to the next step.
		No	Recharge or replace the battery, then go to Step 6. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5]1.)
3	**INSPECT GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) • Is the generator normal?	Yes	Go to the next step.
		No	Replace the generator, then go to Step 6. (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].)
4	**INSPECT INSTRUMENT CLUSTER CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the instrument cluster connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 6.

Step	Inspection		Action
5	**VERIFY INSTRUMENT CLUSTER POWER SUPPLY VOLTAGE** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Display the PID VPWR using the M-MDS. (See PID/DATA MONITOR INSPECTION [INSTRUMENT CLUSTER].) • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the METER1 10 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between METER1 10 A fuse and instrument cluster terminal 1S. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between IG1 relay and instrument cluster terminal 1S. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
6	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the instrument cluster using the M-MDS. (See CLEARING DTC [INSTRUMENT CLUSTER].) • Retrieve the instrument cluster DTCs using the M-MDS. (See DTC INSPECTION [INSTRUMENT CLUSTER].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.) Go to the next step.
		No	Go to the next step.
7	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [INSTRUMENT CLUSTER].)
		No	DTC troubleshooting completed.

With i-stop, Without i-ELOOP

Description	Low power supply voltage input to instrument cluster
Detection condition	• Instrument cluster power supply circuit voltage of **less than 10V** is detected.
Fail-safe	Not applicable

Description	Low power supply voltage input to instrument cluster
Possible cause	• DTCs are stored in the PCM. • Battery malfunction • Generator malfunction • Instrument cluster connector or terminal malfunction • Instrument cluster power supply circuit malfunction — Short to ground in wiring harness between METER1 10 A fuse and instrument cluster terminal 1S — METER1 10 A fuse malfunction — Open circuit in wiring harness between IG1 relay and instrument cluster terminal 1S • Ignition relay (IG1_STAB) malfunction • DC-DC converter connector or terminal malfunction • Wiring harness between DC-DC converter terminal K and instrument cluster terminal 1I malfunction — Short to ground in wiring harness between the following terminals: • METER2 7.5 A fuse—Ignition relay (IG1_STAB) terminal D • Ignition relay (IG1_STAB) terminal C—Instrument cluster terminal 1I — METER2 7.5 A fuse malfunction • Wiring harness between DC-DC converter terminal K and instrument cluster terminal 1I malfunction — METER2 7.5 A fuse malfunction — Open circuit in wiring harness between the following terminals: • DC-DC converter terminal K—Ignition relay (IG1_STAB) terminal D • Ignition relay (IG1_STAB) terminal C—Instrument cluster terminal 1I • DC-DC converter malfunction • Instrument cluster malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY PCM DTCs** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is the battery normal?	Yes	Go to the next step.
		No	Recharge or replace the battery, then go to Step 11. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

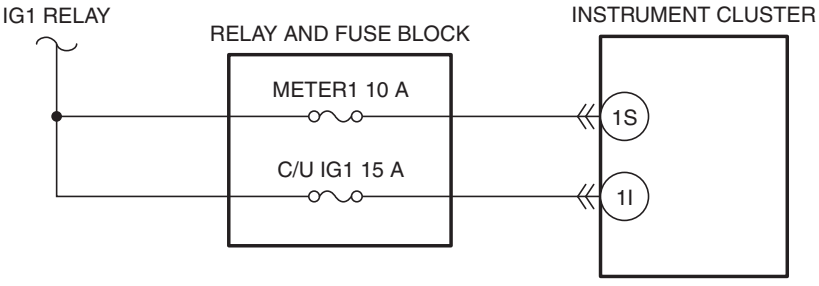
Step	Inspection		Action
3	**INSPECT GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) • Is the generator normal?	Yes	Go to the next step.
		No	Replace the generator, then go to Step 11. (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].)
4	**INSPECT INSTRUMENT CLUSTER CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the instrument cluster connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 11.
5	**VERIFY INSTRUMENT CLUSTER POWER SUPPLY VOLTAGE** • Always reconnect all disconnected connectors. • Reconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Display the PID VPWR using the M-MDS. (See PID/DATA MONITOR INSPECTION [INSTRUMENT CLUSTER].) • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the METER1 10 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector METER1 10 A fuse and instrument cluster terminal 1S. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between IG1 relay and instrument cluster terminal 1S. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 11.
6	**INSPECT IGNITION RELAY (IG1_STAB)** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Remove the ignition relay (IG1_STAB). (See RELAY LOCATION.) • Inspect the ignition relay (IG1_STAB). (See RELAY INSPECTION.) • Is the ignition relay (IG1_STAB) normal?	Yes	Go to the next step.
		No	Replace the ignition relay (IG1_STAB), then go to Step 11. (See RELAY LOCATION.)

Step	Inspection		Action
7	**INSPECT DC-DC CONVERTER CONNECTOR CONDITION** • Disconnect the DC-DC converter connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 11.
8	**INSPECT WIRING HARNESS BETWEEN DC-DC CONVERTER AND INSTRUMENT CLUSTER FOR SHORT TO GROUND** • Verify that the DC-DC converter connector is disconnected. • Disconnect the instrument cluster connector. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Ignition relay (IG1_STAB) terminal D — Instrument cluster terminal 1I • Is there continuity?	Yes	Inspect the METER2 7.5 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between following: • METER2 7.5 A fuse—Ignition relay (IG1_STAB) terminal D • Ignition relay (IG1_STAB) terminal C—Instrument cluster terminal 1I **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. Go to Step 11.
		No	Go to the next step.
9	**INSPECT WIRING HARNESS BETWEEN DC-DC CONVERTER AND INSTRUMENT CLUSTER FOR OPEN CIRCUIT** • Verify that the DC-DC converter and instrument cluster connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — DC-DC converter terminal K—Ignition relay (IG1_STAB) terminal D — Ignition relay (IG1_STAB) terminal C—Instrument cluster terminal 1I • Is there continuity?	Yes	Go to the next step.
		No	Inspect the METER2 7.5 A fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between following: • DC-DC converter terminal K—Ignition relay (IG1_STAB) terminal D • Ignition relay (IG1_STAB) terminal C—Instrument cluster terminal 1I **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 11.
10	**INSPECT DC-DC CONVERTER** • Inspect the DC-DC converter. (See DC-DC CONVERTER INSPECTION [WITHOUT i-ELOOP].) • Is the DC-DC converter normal?	Yes	Go to the next step.
		No	Replace the DC-DC converter, then go to the next step. (See DC-DC CONVERTER REMOVAL/INSTALLATION [WITHOUT i-ELOOP].)

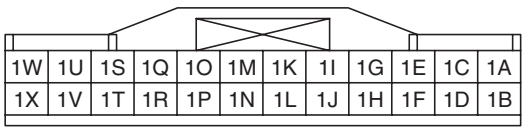
Step	Inspection		Action
11	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the instrument cluster using the M-MDS. (See CLEARING DTC [INSTRUMENT CLUSTER].) • Retrieve the instrument cluster DTCs using the M-MDS. (See DTC INSPECTION [INSTRUMENT CLUSTER].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.) Go to the next step.
		No	Go to the next step.
12	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [INSTRUMENT CLUSTER].)
		No	DTC troubleshooting completed.

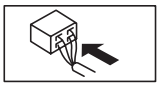
Without i-stop or with i-ELOOP

Description	Low power supply voltage input to instrument cluster
Detection condition	• Instrument cluster power supply circuit voltage of **less than 10V** is detected.
Fail-safe	Not applicable
Possible cause	• DTCs are stored in the PCM. • Battery malfunction • Generator malfunction • Instrument cluster connector or terminal malfunction • Instrument cluster power supply circuit malfunction — Short to ground in wiring harness between METER1 10 A fuse and instrument cluster terminal 1S • METER1 10 A fuse—Instrument cluster terminal 1S • C/U IG1 15 A fuse—Instrument cluster terminal 1I — METER1 10 A fuse malfunction — C/U IG1 15 A fuse malfunction — Open circuit in wiring harness between IG1 relay and instrument cluster terminal 1S • IG1 relay—Instrument cluster terminal 1S • IG1 relay—Instrument cluster terminal 1I • Instrument cluster malfunction



INSTRUMENT CLUSTER
WIRING HARNESS-SIDE CONNECTOR





Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY PCM DTCs** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**INSPECT BATTERY** • Inspect the battery. (See BATTERY INSPECTION.) • Is the battery normal?	Yes	Go to the next step.
		No	Recharge or replace the battery, then go to Step 6. (See BATTERY RECHARGING.) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3	**INSPECT GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-D 1.5)].) • Is the generator normal?	Yes	Go to the next step.
		No	Replace the generator, then go to Step 6. (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See GENERATOR REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
4	**INSPECT INSTRUMENT CLUSTER CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the instrument cluster connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 6.

Step	Inspection		Action
5	**VERIFY INSTRUMENT CLUSTER POWER SUPPLY VOLTAGE** • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at the following terminals (wiring harness-side): — Instrument cluster terminal 1S — Instrument cluster terminal 1I • Is the voltage **B+**?	Yes	Go to the next step.
		No	Inspect the METER1 10 A fuse and C/U IG1 15A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between the following: • METER1 10 A fuse—Instrument cluster terminal 1S • C/U IG1 15 A fuse—Instrument cluster terminal 1I **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between the following: • IG1 relay—Instrument cluster terminal 1S • IG1 relay—Instrument cluster terminal 1I **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
6	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the instrument cluster using the M-MDS. (See CLEARING DTC [INSTRUMENT CLUSTER].) • Retrieve the instrument cluster DTCs using the M-MDS. (See DTC INSPECTION [INSTRUMENT CLUSTER].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.) Go to the next step.
		No	Go to the next step.
7	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [INSTRUMENT CLUSTER].)
		No	DTC troubleshooting completed.